

Decoupling of the Onset of Anharmonicity between a Protein and Its Surface Water around 200 K

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Abstract

The protein dynamical transition at ~ 200 K, where the biomolecule transforms from a harmonic, non-functional form to an anharmonic, functional state, has been thought to be slaved to the thermal activation of dynamics in its surface hydration water. Here, by selectively probing the dynamics of protein and hydration water using elastic neutron scattering and isotopic labelling, we found that the onset of anharmonicity in the two components around 200 K are decoupled. The one in protein is an intrinsic transition, whose characteristic temperature is independent of the instrumental resolution time, but varies with the biomolecular structure and the amount of hydration, while the one of water is merely a resolution effect.

eLife assessment

The study answers the **important** question of whether the conformational dynamics of proteins are slaved by the motion of solvent water or are intrinsic to the polypeptide. The results from neutron scattering experiments, involving isotopic labelling, carried out on a set of four structurally different proteins are **convincing**, showing that protein motions are not coupled to the solvent. A strength of this work is the study of a set of proteins using spectroscopy covering a range of resolutions. A minor weakness is the limited description of computational methods and analysis of data. The work is of broad interest to researchers in the fields of protein biophysics and biochemistry.

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Introduction

It is well established that the internal dynamics of a protein is crucial for its functions, including allosteric conformational changes (1), ligand binding (2) and enzymatic reactions (3). In particular, hydrated proteins exhibit a dynamical transition around 200 K, across which the slope of the temperature dependence of the atomic displacements changes significantly and the biomolecule transforms from a rigid, harmonic state, to a flexible, anharmonic form (4-12). Although exceptions have been reported (50), the dynamical transition has been linked to the thermal onset of function in a number of proteins, e.g. myoglobin (13), ribonuclease (6), elastase (14) and bacteriorhodopsin (15), all of which become inactive below the dynamical transition temperature. The dynamical transition of protein has garnered various explanations. One theory suggests it's due to the behavior of water in the hydration shell, transitioning from rigid to fluid at certain temperatures, thus influencing protein flexibility (7,12,16-19). Another theory considers the transition as an inherent property of the protein, where thermal energy allows the protein to access a wider range of conformations (21).

A prevailing scenario is that the internal dynamics of the protein is slaved to the motion of the surrounding hydration water (7,12,16-19), and thus the protein dynamical transition results from the changes in the dynamics of the hydration water with temperature (5,7,12,17,20). This scenario is indirectly supported by the experimental finding that the presence of the protein dynamical transition requires a minimum amount of hydration water, ~ 0.2 gram water/gram protein (4,8). Further support comes from the results of all-atom molecular dynamics simulations, suggesting that it is the activation of the translational motions of surface water molecules around 200 K that leads to the dynamical transition in the underlying protein (5,7,20).

This 'slaving' scenario can be examined directly by an experiment using isotopic labelling in combination with elastic neutron scattering methods (7,21). Neutrons are highly sensitive to hydrogen atoms as their incoherent scattering cross section is an order of magnitude higher than the incoherent and coherent scattering cross sections of other elements (22-24). Thus, neutron signals collected on an ordinary protein powder hydrated in D₂O reflect the dynamics of the protein while signals from the perdeuterated sample in H₂O inform about the motion of water. The experimental results derived from this combined approach are, however, inconsistent (7,21,25). Measurements performed on perdeuterated maltose binding protein hydrated in H₂O revealed a harmonic-to-anharmonic transition for hydration water taking place at the same temperature as that of the underlying protein (7). In contrast, a similar experiment on perdeuterated green fluorescence protein showed that the anharmonic onset in hydration water occurs at a lower temperature than that of the protein (21). More recent measurements on lysozyme hydrated in both D₂O and H₂O found that the transition temperature of protein and water coincided when examining their atomic displacements at 1 ns, but took place at different temperatures when changing the explored time scale to 3 ns (25). Therefore, there remains an unanswered question concerning whether the transition in dynamics of protein around 200 K is indeed coupled to that of the hydration water, whose resolution is of fundamental importance to understand the mechanism governing the nature of their interaction.

To address this, it requires a systematic measurement of the temperature dependence of atomic displacements of the protein and its surface water separately, as a function of hydration levels, *h* (gram water/gram protein), and at different time scales (instrument resolutions). Here, we performed elastic neutron scattering experiments on a number of protein powders hydrated in D₂O and on the perdeuterated counterparts hydrated in H₂O, to track the dynamics of protein and hydration water independently. Moreover, using a range of neutron instruments with distinct resolutions, we tested the effect of the explored time scales on the dynamics of the two

components. Four globular proteins with different secondary and tertiary structures (see Fig.S1 and Table S1) were studied here. We found that the onset temperature (T_{on}) of the protein dynamical transition varies with both biomolecular structure and hydration level, but is independent of the instrumental resolution time. Conversely, T_{on} of the hydration water is insensitive to both the protein structure and the level of hydration, but solely determined by the instrument resolution. Therefore, the dynamical transition of the protein is decoupled from the onset of anharmonic dynamics of its hydration water around 200 K. The onset in water cannot be assigned to a physical transition, but to a resolution effect. In contrast, the protein dynamical transition is an intrinsic change in the dynamics of the biomolecules. Complementary differential scanning calorimetry (DSC) measurements revealed a step-like change in the heat flow around the transition temperature of the protein, similar to the glass transition observed in polymers. This suggests that the dynamical transition in the protein results from a similar process involving the freezing of the structural relaxation of the protein molecules beyond equilibrium.

Results

Elastic Neutron Scattering Experiments

The quantity measured in the neutron experiment is the elastic intensity, i.e. the intensity of the elastic peak in the dynamic structure factor, $S(q, \Delta t)$, where q is the scattering wave vector and Δt is the resolution time of the neutron spectrometer. $S(q, \Delta t)$ is an estimate of the average amplitude of the atomic motions up to Δt (11,22). Three neutron backscattering spectrometers were chosen to cover a wide range of timescales; HFBS at the NIST Center for Neutron Scattering, USA, DNA at the Materials and Life Science Experimental Facility at J-PARC in Japan, and OSIRIS at the ISIS Neutron and Muon Facility, UK. The instrumental energy resolutions are 1 μeV , 13 μeV , 25.4 μeV and 100 μeV , corresponding to timescales of ~ 1 ns, ~ 80 ps, ~ 40 ps, and ~ 10 ps, respectively. Four globular proteins were investigated, myoglobin (MYO), cytochrome P450 (CYP), lysozyme (LYS), and green fluorescent protein (GFP), the detailed structural features of which are presented in Fig. S1 and Table S1. For simplicity, the hydrogenated and perdeuterated proteins are noted as H-protein and D-protein, respectively. Details of the sample preparation and neutron experiments are provided in Supplemental Material (SM).

Dynamics of protein

Figs. 1a to g show the temperature dependence of $S(q, \Delta t)$ collected on H-LYS, H-MYO and H-CYP in dry and hydrated state with D_2O measured by neutron spectrometers of different resolutions, Δt . Since the measurements were performed on H-protein in D_2O , the signals reflect the dynamics of the proteins. A clear deviation can be seen in the temperature dependence of $S(q, \Delta t)$ for the hydrated protein from that of the dry powder, which is defined as the onset temperature, T_{on} (8,12,25-27), of the protein dynamical transition. The advantage of such definition is that it highlights the effect of hydration on the anharmonic dynamics in proteins while removing the contribution from the local side groups, e.g., methyl groups, whose motions are hydration independent (11,28). As shown in Ref. (12,29), the activation temperature of the rotations of methyl group varies with the instrument resolution, which will cloud the present analysis. Two important conclusions can be drawn from **Fig.1** (T_{on} is summarized in Table S5). (i) T_{on} is distinct for each protein, LYS (213 K), MYO (198 K) and CYP (228 K), and (ii) it is independent of the timescale explored even though the resolutions of the neutron spectrometers differ by orders of magnitude. Using the same set of data, we also analyzed the temperature dependence of the mean squared atomic displacements, $\langle x^2(\Delta t) \rangle$ (see details in SM and results in Fig. S2) and obtained similar conclusions. We further calculated T_{on} of H-protein in D_2O in the q -range from 0.45-0.9 $\text{\AA}^{-1}$ and 1.1-1.75 $\text{\AA}^{-1}$. As shown in Table S3-S4, the q -range does not alter the T_{on} of proteins.

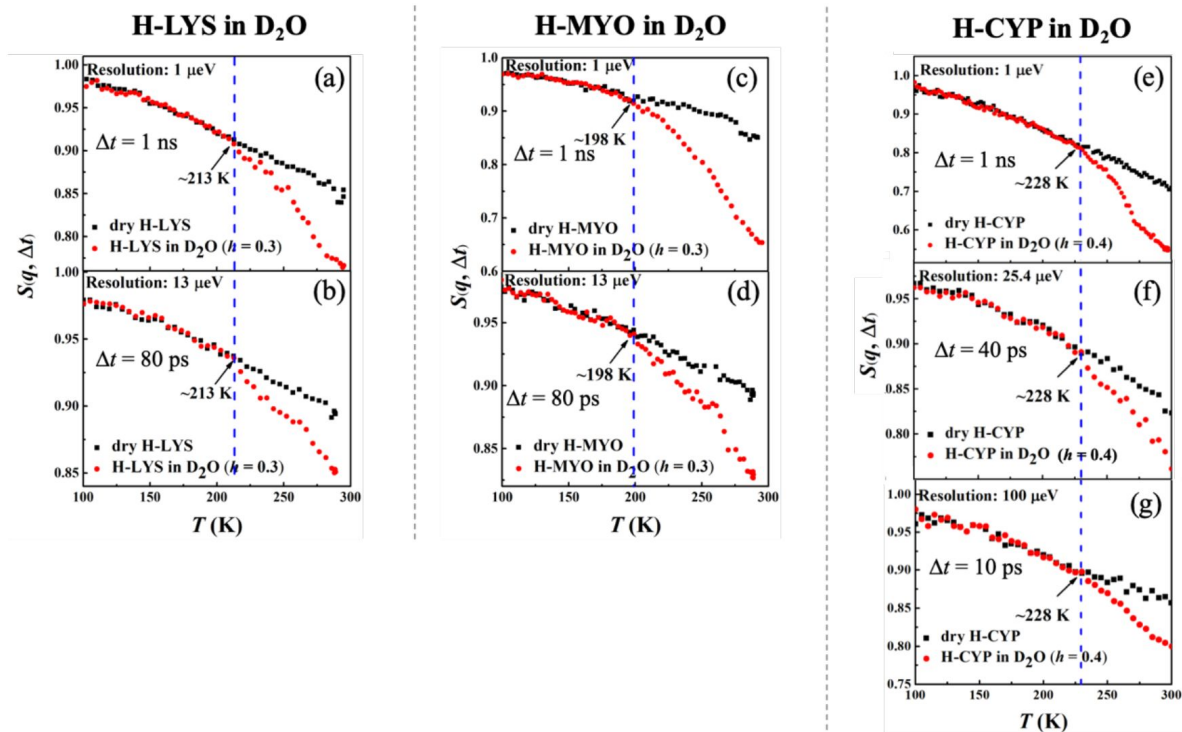


Figure 1.

Resolution dependence of the onset of protein dynamical transition. Neutron spectrometers with different resolutions (1, 13, 25.4 and 100 μeV) were applied. Elastic intensity $S(q, \Delta t)$ of (a, b) dry H-LYS and H-LYS in D_2O at $h = 0.3$, (c, d) dry H-MYO and H-MYO in D_2O at $h = 0.3$, and (e-g) dry H-CYP and H-CYP in D_2O at $h = 0.4$. All the experimental $S(q, \Delta t)$ are normalized to data measured at ~ 10 K and summed over values of q ranging from 0.45 to $1.75 \text{ \AA}^{-1}$. The dashed lines in each figure identify the onset temperatures of the transition, T_{on} , where the neutron data of the hydrated system deviate from the dry form. The same analyses are used in **Figs. 2** to **4** and S2 to S5.

These findings suggest that the dynamical transition in the protein is an intrinsic property of the hydrated biomolecule, and it depends on the structure and chemistry of the protein concerned. Our results are consistent with Ref. (12), which demonstrated that T_{on} in both protein and polypeptide is independent of the resolution of the neutron spectrometer, if one carefully removes the contributions from methyl rotations and vibrations to $\langle x^2(\Delta t) \rangle$ measured by elastic neutron scattering. Additionally, Ref. (25) showed that, as compared to the dry form, the D_2O -hydrated lysozyme presents an approximately resolution-independent T_{on} , again in agreement with our findings.

Fig. 2 compares the temperature dependence of $S(q, \Delta t)$ measured on H-CYP and H-LYS in D_2O at different hydration levels, h (T_{on} is summarized in Table S6). Evidently T_{on} of the protein increases from 228 to 248 K when reducing h from 0.4 to 0.2 (**Figs. 2a**) by using the neutron instrument with $\Delta t = 1$ ns. A similar hydration dependence of T_{on} is also observed when we replot the neutron data measured on H-LYS hydrated in D_2O reported in Ref. (26) (**Fig. 2c**). It can be found that T_{on} of lysozyme changes systematically from 195 to 225 K, when decreasing h from 0.45 to 0.18. Similar conclusion can be obtained when we analyzed $\langle x^2(\Delta t) \rangle$ (see Fig. S3). The dynamical transition temperature in lipid membranes is higher when the membrane is dry (51). We also studied the secondary structure content and tertiary structure of CYP protein at different hydration levels ($h = 0.2$ and 0.4) through molecular dynamics simulation. As shown in Table S2 and Fig. S6, the extent of hydration does not alter the protein secondary structure content and overall packing. Thus, this result suggests that water molecules have more influence on protein dynamics than on protein structure.

The results from the neutron scattering experiments suggest that the dynamical transition in proteins is an intrinsic property of the biomolecule and strongly depends on the amount of water surrounding it. Such an intrinsic transition can result either from a critical phase transition, e.g., water to ice (30,31), or from freezing of the structural relaxation of the system beyond the equilibrium time ($\sim 100 - 1000$ s) of the experiment, in analogy to the glass transition in polymers from rubbery state to the glass form (32-34). Both of them will significantly increase the mechanical modulus of the material and suppress the atomic displacements at the fast time scales (pico-to-nanosecond) probed by the neutron spectrometers (30-34) like those used in this work. To explore the microscopic nature of the protein dynamical transition, we performed differential scanning calorimetry (DSC) measurements on CYP at dry, $h = 0.2$ and 0.4 . As illustrated by **Fig. 2b**, H-CYP at $h = 0.2$ and $h = 0.4$ exhibit a step-like transition in the heat flow at 245 K and 225 K, respectively, while no such transition is observed in dry H-CYP. Such step-like transition in heat flow is normally defined as the glass transition in polymers (35,36).

For simplicity, the step-like transition identified by DSC is noted as T_{DSC} . When comparing **Figs 2b** with **Fig. 2a**, one can find that the values of T_{DSC} approximate those of T_{on} probed by neutrons. T_{DSC} of hydrated myoglobin was reported by literature to be 190 K (37), which is again in good agreement with the value of T_{on} in **Fig. 1**. More importantly, T_{DSC} and T_{on} present the same hydration dependence, i.e., both increase with decrease of h (see **Figs. 2a** and **2b**). Therefore, we can conclude that the onset of anharmonicity around 200 K in proteins measured by neutron scattering as shown in **Figs. 1** and **2** results from the freezing of the structural relaxation of the biomolecule beyond the equilibrium when cooling the system below T_{DSC} , similar to the glass transition in polymers. Similar interpretation has also been suggested in Ref. (38).

As the time scale probed by neutron spans from pico-to nanoseconds, it is too fast to allow us to directly “see” structural relaxations of the protein around T_{on} . However, “freezing” of the structural relaxation beyond the equilibrium time ($\sim 100 - 1000$ s), i.e., the measurement time of neutron experiments at each temperature, will turn the system into a “frozen” solid form, which can significantly suppress the fast dynamics measured by neutron and cause the transition probed (32-34,38). Moreover, water can be considered here as lubricant or plasticizer which

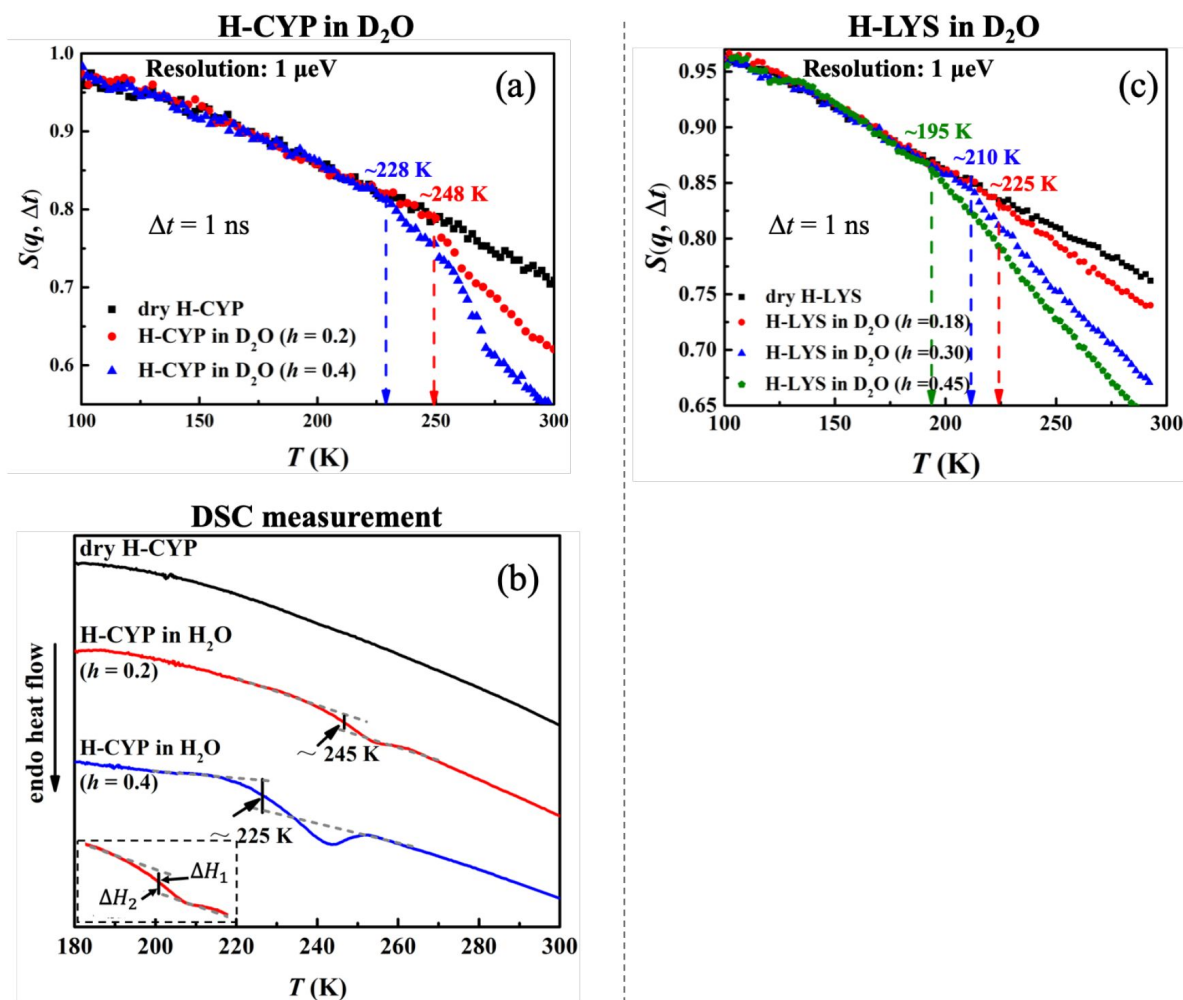


Figure 2.

Hydration dependence of the onset of protein dynamical transition. $S(q, \Delta t)$ of (a) dry H-CYP and H-CYP in D₂O at $h = 0.2$ and 0.4 and (c) dry H-LYS and H-LYS in D₂O at $h = 0.18, 0.30$ and 0.45 , all measured using HFBS with the instrumental resolution of $1 \mu\text{eV}$. All the Data in (c) were replotted from Ref. (26). (b) DSC curves obtained for dry H-CYP and H-CYP in water at $h = 0.2$ and 0.4 . T_{DSC} is defined as the midpoint between two heat flow baselines, where (35,36,49).

facilitates the motion of the biomolecule (28,39,40). As widely observed in polymeric systems (41–43), adding water as plasticizer will significantly reduce the glass transition temperature of the polymers. This rationalizes the hydration effect on T_{DSC} and T_{on} , both decreasing with increase of h .

Dynamics in hydration water

Figs. 3a–b and **c–e** show the temperature dependence of $S(q, \Delta t)$ measured on perdeuterated GFP and CYP in dry and H_2O -hydrated forms. In these samples, the measured signal reflects primarily the motions of water molecules. Two important observations arise from the data. Firstly, T_{on} of hydration water for these two proteins strongly depends on the resolution of the spectrometer, increasing drastically from 200 K to 250 K when reducing Δt from 1 ns to 10 ps. Similar conclusions are obtained when we analyzed the temperature dependence of $\langle x^2(\Delta t) \rangle$ (see Fig. S4). The observation of a dependence of T_{on} on Δt is typical for a thermally activated process, which occurs when the characteristic relaxation time becomes comparable to the instrumental resolution, and the relaxation process is said to enter the time window of the instrument (29,44). In this case, it means that the relaxation time, τ , of hydration water is 10 ps at 250 K, 40 ps at 234 K and 1 ns at 200 K. Assuming an Arrhenius-type process, $\tau = \tau_0 \exp(\frac{\Delta U}{k_B T})$, the energy barrier ΔU can be estimated to be ~ 38 kJ/mol. Our results thus demonstrate that the anharmonic onset of the hydration water is in fact not a real physical transition but merely a resolution effect. It occurs as the relaxation time τ of water, which varies continuously with temperature, happens to cross the instrument resolution, Δt , on the pico-to-nanosecond time scales at T_{on} . Our findings agree with reports from dielectric measurements, the signal of which is highly sensitive to the rotation of hydration water (45,46). They showed a smooth temperature dependence of the characteristic relaxation time in the range from 170 K to 250 K without any sudden changes (45,46). Moreover, our data also agree with Ref. (47), which demonstrated that the characteristic relaxation time of protein-surface water, measured on H_2O -hydrated perdeuterated C-phycocyanin by quasi-elastic neutron scattering, changes smoothly over temperature without any disruptions around the dynamical transition temperature of the protein. Secondly, the onset temperature of the hydration water is independent of the protein structure when Δt is fixed, since the values of both GFP and CYP are identical.

Furthermore, the hydration dependence of the anharmonic onset of the water is presented in **Fig. 4**, which shows that T_{on} remains constant with h as long as the instrument resolution is fixed. This behavior is drastically different from that of the protein (**Fig. 2**). The same conclusions can be obtained when analyzing $\langle x^2(\Delta t) \rangle$ (see Figs. S5).

Conclusion and Discussion

By combining elastic neutron scattering with isotopic labelling, we have been able to probe the dynamics of the protein and surface water separately, as a function of temperature, protein structural composition, hydration level and time scale. We found that the anharmonic onsets of the two components around 200 K are clearly decoupled and different in origin. The protein shows an intrinsic transition, whose temperature depends on the structure of the protein and the hydration level, and not on the instrument used to measure it. It has a thermodynamic signature similar to the glass transition in polymers as confirmed by DSC, and thus can be assigned to the freezing of the structural relaxation of the protein beyond the experimental equilibrium time (100 to 1000 s). In contrast, the temperature at which the onset of anharmonicity happens in the hydration water is given by the instrument resolution, independent of both the biomolecular structure and the level of hydration.

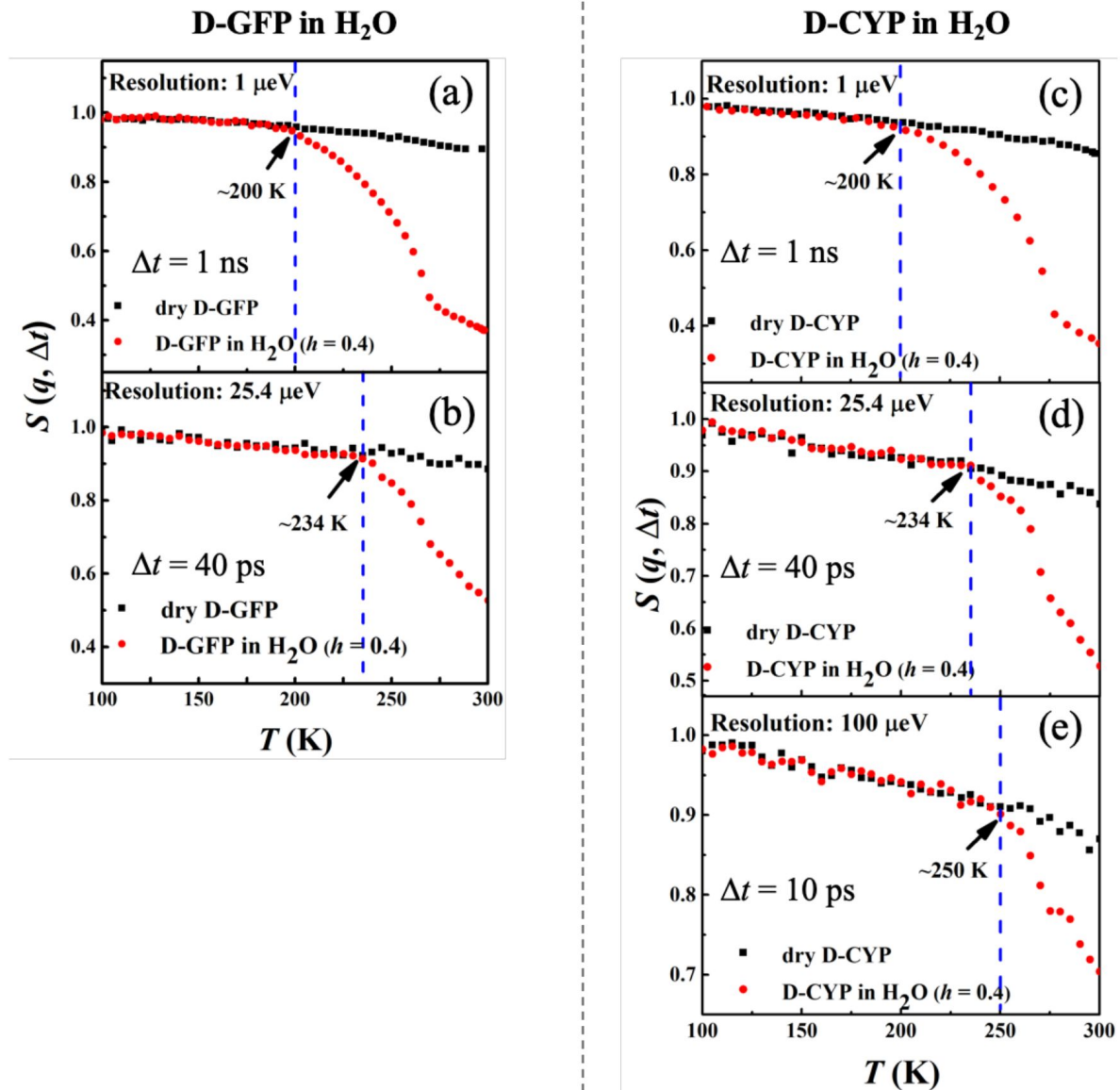


Figure 3.

Resolution dependence of the anharmonic onset of hydration water. Neutron spectrometers with different resolutions (1, 25.4 and 100 μeV) were applied. $S(q, \Delta t)$ of (a, b) dry D-GFP and D-GFP in H₂O at $h = 0.4$, and (c-e) dry D-CYP and D-CYP in H₂O at $h = 0.4$.

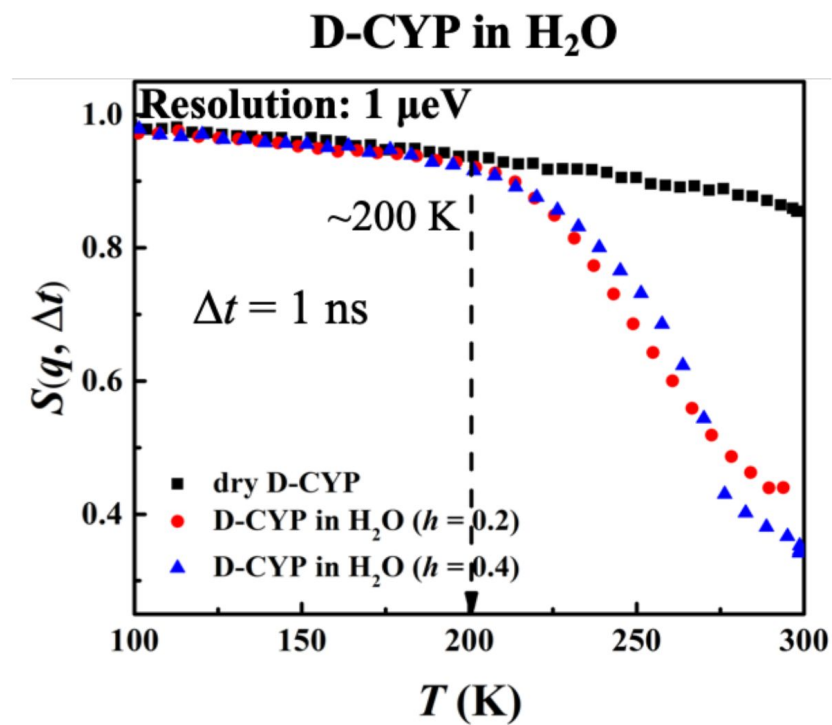


Figure 4.

Hydration dependence of the anharmonic onset of hydration water. $S(q, \Delta t)$, for dry D-CYP and D-CYP in H₂O at $h = 0.2$ and 0.4 , measured using HFBS neutron instrument with an energy resolution of $1\text{ }\mu\text{eV}$.

Based on our findings, we can infer that, in some cases, the dynamical transition of a protein can coincide with the anharmonic onset of its surface water if one characterizes the system using a single neutron instrument with a fixed resolution. But such coincidence will be torn apart if the measurements were conducted by using instruments of different resolutions or at different amounts of hydration, such as in the present work. This rationalizes the seemingly contradictory results reported in the literature (7 [↗](#), 21 [↗](#), 25 [↗](#)).

The protein dynamical transition has long been thought to connect to the thermal onset of the functionality of the biomolecule. Our experiments suggest that this transition in protein is an intrinsic property of the hydrated protein that its structural relaxation is activated upon heating above the onset temperature. This structural relaxation might be associated with conformational jumps of the biomolecules among different functional states, such as the states with the ligand-binding pocket being opened or closed. Unfreezing of the protein structural relaxation might facilitate these conformational jumps, turning on its functionality. However, as revealed by Ref (48 [↗](#)), the denatured form of lysozyme also exhibits a dynamical transition, similar to that seen in its folded native form. Additionally, the dynamical transition also can be found in the mixture of amino acids (12 [↗](#)). Hence, one can argue that the activation of the structural relaxation of the biomolecule above the dynamical transition temperature is a necessary but insufficient condition for the protein to function, as the latter also requires the biomolecule assuming the correctly folded 3-dimensional structure. The findings in this work help further the understanding of the microscopic mechanism governing the dynamics in proteins and their hydration water, as well as their interactions at the cryogenic temperature. More importantly, we demonstrated that the protein dynamical transition is a real transition, connecting to unfreezing of the biomolecular structural relaxation, which could be crucial for activating the function.

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Editors

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Reviewer #1 (Public Review):

Summary:

Zheng et al. study the 'glass' transitions that occurs in proteins at ca. 200K using neutron diffraction and differential isotopic labeling (hydrogen/deuterium) of the protein and solvent. To overcome limitations in previous studies, this work is conducted in parallel with 4 proteins (myoglobin, cytochrome P450, lysozyme and green fluorescent protein) and experiments were performed at a range of instrument time resolutions (1ns - 10ps). The author's data looks compelling, and suggests that transitions in the protein and solvent behavior are not coupled and contrary to some previous reports, the apparent water transition temperature is a 'resolution effect'; i.e. instrument response limited. This is likely to be important in the field, as a reassessment of solvent 'slaving' and the role of the hydration shell on protein dynamics should be reassessed in light of these findings.

Strengths:

The use of multiple proteins and instruments with a range of energy resolution/ timescales.

Weaknesses:

The paper could be organised to better allow the comparison of the complete dataset collected.

The extent of hydration clearly influences the protein transition temperature. The authors suggest that "water can be considered here as lubricant or plasticizer which facilitates the motion of the biomolecule." This may be the case, but the extent of hydration may also alter the protein structure.

<https://doi.org/10.7554/eLife.95665.2.sa2>

Reviewer #2 (Public Review):

Summary:

The manuscript entitled "Decoupling of the Onset of Anharmonicity between a Protein and Its Surface Water around 200 K" by Zheng et al. presents a neutron scattering study trying to elucidate if at the dynamical transition temperature water and protein motions are coupled. The origin of the dynamical transition temperature is highly debated since decades and specifically its relation to hydration.

Strengths:

The study is rather well conducted, with a lot of efforts to acquire the perdeuterated proteins, and some results are interesting.

Weaknesses:

The MD data presented appears to be missing description of the methods used. If these data support the authors claim that different levels of hydration do not affect the protein structure, careful analysis of the MD simulation data should be presented that show the systems are properly equilibrated under each condition. Additionally, methods are needed to describe the MD parameters and methods used, and for how long the simulations were run.

<https://doi.org/10.7554/eLife.95665.2.sa1>

Author response:

The following is the authors' response to the original reviews.

eLife assessment:

The study answers the important question of whether the conformational dynamics of proteins are slaved by the motion of solvent water or are intrinsic to the polypeptide. The results from neutron scattering experiments, involving isotopic labelling, carried out on a set of four structurally different proteins are convincing, showing that protein motions are not coupled to the solvent. A strength of this work is the study of a set of proteins using spectroscopy covering a range of resolutions, however, it suffers from some scholarly shortcomings and limited discussion of results. The work is of broad interest to researchers in the fields of protein biophysics and biochemistry.

Reply 1: We thank the editors and reviewers for the positive and encouraging comments.

Reviewer #1 (Public Review):

Summary:

Zheng et al. study the 'glass' transitions that occur in proteins at ca. 200K using neutron diffraction and differential isotopic labeling (hydrogen/deuterium) of the protein and

solvent. To overcome limitations in previous studies, this work is conducted in parallel with 4 proteins (myoglobin, cytochrome P450, lysozyme, and green fluorescent protein) and experiments were performed at a range of instrument time resolutions (1ns - 10ps). The author's data looks compelling, and suggests that transitions in the protein and solvent behavior are not coupled and contrary to some previous reports, the apparent water transition temperature is a 'resolution effect'; i.e. instrument response is limited. This is likely to be important in the field, as a reassessment of solvent 'slaving' and the role of the hydration shell on protein dynamics should be reassessed in light of these findings.

Strengths:

The use of multiple proteins and instruments with a range of energy resolution/ timescales.

Reply 2: We thank the reviewer for highlighting our key findings.

Weaknesses:

The paper could be organised to better allow the comparison of the complete dataset collected. The extent of hydration clearly influences the protein transition temperature. The authors suggest that "water can be considered here as lubricant or plasticizer which facilitates the motion of the biomolecule." This may be the case, but the extent of hydration may also alter the protein structure.

Reply 3: Following the reviewer's suggestion, we studied the secondary structure content and tertiary structure of CYP protein at different hydration levels ($h = 0.2$ and 0.4) through molecular dynamics simulation. As shown in Table S2 and Figure S6, the extent of hydration does not alter the protein secondary structure content and overall packing. Thus, this result also suggests that water molecules have more influence on protein dynamics than on protein structure.

Reviewer #2 (Public Review):

Summary:

The manuscript entitled "Decoupling of the Onset of Anharmonicity between a Protein and Its Surface Water around 200 K" by Zheng et al. presents a neutron scattering study trying to elucidate if at the dynamical transition temperature water and protein motions are coupled. The origin of the dynamical transition temperature has been highly debated for decades, specifically its relation to hydration.

Strengths:

The study is rather well conducted, with a lot of effort to acquire the perdeuterated proteins, and some results are interesting.

Reply 4: We thank the reviewer for highlighting our key findings.

Weaknesses:

The present work could certainly contribute some arguments, but I have the feeling that not all known facts are properly discussed.

The points the authors should carefully discuss are the following:

(1) Daniel et al. (10.1016/S0006-3495(98)77694-5) have shown that enzymes can be functional below the dynamical transition temperature which is at odds with some of the

claims of the authors.

Reply 5: Following the reviewer's suggestion, we added the following paragraph into the Introduction into the revised main text.

"Although exceptions have been reported (Biophys. J. 1998, 75, 2504.), the dynamical transition has been linked to the thermal onset of function in a number of proteins, e.g, myoglobin (Biochemistry, 1975, 14, 5355-5373), ribonuclease (Nature, 1992, 357, 423-424.), elastase (Biochemistry, 1994, 33, 9285-9293.) and bacteriorhodopsin (PNAS, 1993, 90, 9668-9672.), all of which become inactive below the dynamical transition temperature."

(2) It is not as easy to say that protonated proteins in D2O reflect protein dynamics while perdeuterated proteins in H2O reflect water dynamics. A recent study by Nidriche et al. (PRX LIFE 2, 013005 (2024)) reveals that H <-> D exchange is much faster than usually assumed and has important consequences for such studies.

Reply 6: For the sample preparation, all the H-proteins were dissolved in D2O to allow full deuterium exchange of all exchangeable hydrogen atoms and then lyophilized for 12 hours to obtain the dry sample. The lyophilized H-protein is then put into a desiccator with D2O, placed in the glove box purged with nitrogen gas, to absorb D2O till the desired hydration level, h (gram water/gram protein). In contrast, the preparation of the deuterated proteins was conducted in the opposite way. The D-proteins were dissolved in H2O to allow full hydrogen exchange of all exchangeable deuterium atoms and then lyophilized for 12 hours to obtain the dry sample. The lyophilized D-protein is then put into a desiccator with H2O to absorb H2O till the desired h. This procedure can avoid H-D exchange during experiments. We added the above methods into the revised SI.

(3) A publication by Jasnin et al. (10.1039/b923878f) on heparin sulfate shows a resolution effect.

Reply 7: Based on the data from Jasnin et al. (10.1039/b923878f), we found that the dynamical transition of heparin sulfate did not exhibit a strong resolution effect. Estimating the dynamical transition of mean square displacement (MSD) for nanosecond motions in all heparan sulfate samples is challenging due to the absence of data on nanosecond motion of HS-dry.

(4) The authors should discuss the impact of the chosen q-range on their findings (see Phys. Chem. Chem. Phys., 2012, 14, 4927-4934, where the authors see a huge effect!).

Reply 8: Following the reviewer's suggestion, we calculated Ton of H-protein in D2O in the q-range from 0.45-0.9 Å⁻¹ and 1.1-1.75 Å⁻¹. The results are summarized in Table S2 and Table S3. As shown in Tables S2-3., the q-range does not alter the Ton of proteins. We added the above results into the revised SI.

(5) The authors underline that the dynamical transition is intrinsic to the protein. However, Cupane et al. (ref 12) have shown that it can also be found in a mixture of amino acids without any protein backbone.

Reply 9: Following the reviewer's suggestion, we added the following discussion into the revised main text.

"Unfreezing of the protein structural relaxation might facilitate these conformational jumps, turning on its functionality. However, as revealed by Ref (Journal of biological physics, 2010, 36, 291-297.), the denatured form of lysozyme also exhibits a dynamical transition, similar to that seen in its folded native form. Additionally, the dynamical transition also can be found in

the mixture of amino acids (Physical Review Letters, 2012, 109, 128102.). Hence, one can argue that the activation of the structural relaxation of the biomolecule above the dynamical transition temperature is a necessary but insufficient condition for the protein to function, as the latter also requires the biomolecule assuming the correctly folded 3-dimensional structure.”

(6) The authors say that they find similar dependences from MSD. They should explain that the MSD is inversely proportional to the summed intensities squared.

Reply 10: Following the reviewer’s suggestion, we added the estimation of mean-squared atomic displacement (MSD) in the revised SI.

“The mean-squared atomic displacement $\langle x^2(\Delta t) \rangle$ was estimated by performing Gaussian approximation, where $S(q, \Delta t) = \exp(-\frac{1}{6} q^2 \langle x^2(\Delta t) \rangle)$. The values of q used for Gaussian fitting ranges from 0.45 to 0.9 Å (Biophys. J. 2006, 91, 2573.).”

(7) A decoupling between water dynamics and membrane dynamics has already been discussed by K. Wood, G. Zaccai et al.

Reply 11: Following the reviewer’s suggestion, we added the discussion in revised main text. “The results from the neutron scattering experiments suggest that the dynamical transition in proteins is an intrinsic property of the biomolecule and strongly depends on the amount of water surrounding it. Such an intrinsic transition can result either from a critical phase transition, e.g., water to ice (PNAS 2007, 104, 18049-18054.; JPCB, 1999, 103, 8036-8050), or from freezing of the structural relaxation of the system beyond the equilibrium time (~100-1000 s) of the experiment, in analogy to the glass transition in polymers from rubbery state to the glass form (Philosophical Magazine, 2004, 84, 1341-1353.; Science, 1995, 267, 1939-1945.; Colloid and Polymer Science, 1995, 273, 413-420.).”

(8) The fact that transition temperature in lipid membranes is higher when the membrane is dry is also well known (A.V. Popova, D.K. Hinch, BMC Biophys. 4, 11 (2011)).

Reply 12: We agree with the reviewer that transition temperature in lipid membranes is higher when the membrane is dry is well known. We cited this work as reference.

(9) The authors should mention the slope (K/min) they used for DSC and discuss the impact of it on the results.

Reply 13: Following the reviewer’s suggestion, we added DSC measurements in revised SI. “DSC measurements were performed by using the METTLER instruments DSC3+. The sample was sealed in a pan of aluminum. An empty pan was used as a reference. All the experiments were carried out in the temperature range from 150 to 300 K with a heating rate of 1 K/min. The heating rate of DSC is the same as neutron experiments.”

(10) In the introduction, the authors should present the different explanations forwarded for the dynamical transition.

Reply 14: Following the reviewer’s suggestion, we added different explanations forwarded for the dynamical transition in the Introduction in revised main text.

“The dynamical transition of protein represents a significant change in the internal mobility of proteins, which has garnered various explanations. One theory suggests it's due to the behavior of water in the hydration shell, transitioning from rigid to fluid at certain temperatures, thus influencing protein flexibility. Another theory considers the transition as an inherent property of the protein, where thermal energy allows the protein to access a wider range of conformations.”

Reviewer #1 (Recommendations For The Authors):

A major strength of the work is the parallel experiments performed on each of the 4 proteins. To allow better comparison of these it would be helpful to present these combined data in relevant figures to make a side-by-side comparison easier. A summary table of Ton (and potentially TDSC) values would also be helpful.

Reply 15: Following the reviewer's suggestion, we summarized the Ton of proteins in Table S5 and Table S6.

The effect of hydration on protein structure should be considered. Alterations in protein secondary and tertiary structure would be expected to alter dynamics and thus could be seen as a change in Ton.

Reply 16: The detailed analysis and discussion are presented in Reply 3.

No uncertainty (error) in Ton values is presented. Could these be estimated from e.g. a comparison of protein Ton values measured under identical sample conditions with different spectrometers?

Reply 17: It would be hard to compare Ton of proteins measured with different spectrometers because different spectrometers have different energy resolutions. For example, the energy resolutions of HFBS, DNA and OSIRIS are 1 μeV , 13 μeV , 25.4 μeV and 100 μeV , respectively.

More detail is needed to correctly describe/define the proteins used for the study - e.g. P450 is a family of enzymes, so which one was used?

Reply 18: We used P450 from *Pseudomonas putida* for the study. The PDB ID is 2ZAX. We added this information in the revised SI.

P450 and myoglobin also have heme cofactors. Were these deuterated as part of the protein preparation?

Reply 19: The heme cofactors were deuterated as part of the protein preparation. For D-protein, all the cell culture for *E.coli* is deuterated.

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